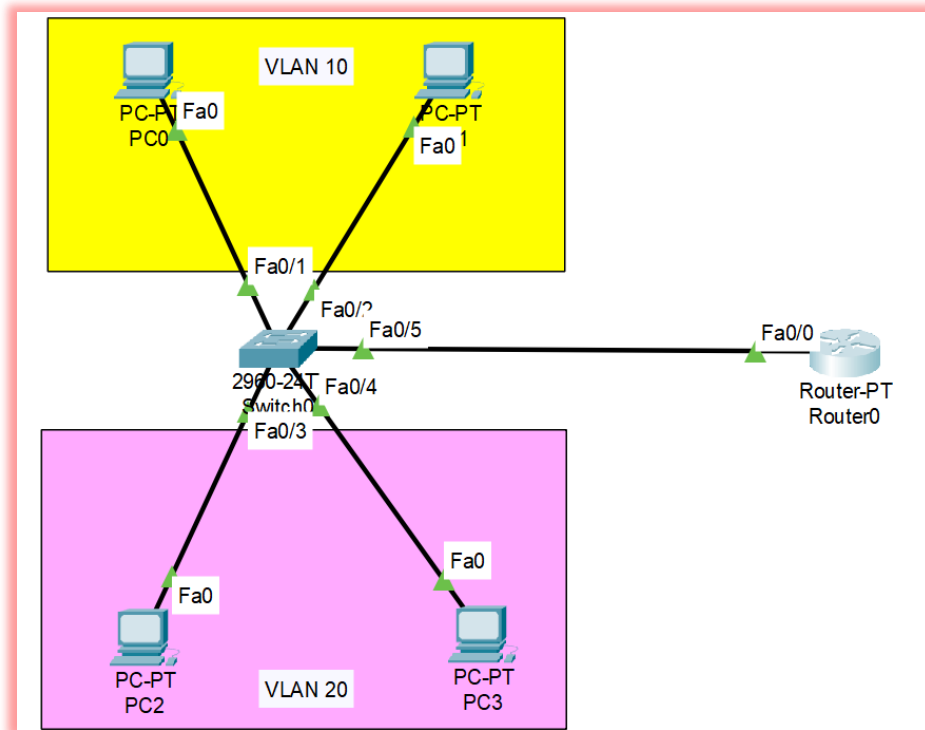


Lab 12 Tasks Report Submission

Task-01

Using the provided network topology, configure Virtual Local Area Networks (VLANs) to segment the network based on department or functional requirements. Assign appropriate VLAN IDs, configure VLAN interfaces on the switches, and ensure proper VLAN membership for all connected devices. Verify inter-VLAN communication through a router or Layer 3 switch if required.



Task-02

At FAST Technologies, the network administrator Nadeem has been asked to redesign the LAN infrastructure to improve security, reduce broadcast traffic, and ensure efficient inter-department communication. The company has three main departments: HR, Finance, and IT, each with its own set of computers and operational needs. Nadeem decides to implement VLANs to logically separate each department and configure Inter-VLAN Routing so that they can still communicate with each other when necessary.

Each department is connected to an access switch (configured with VLANs), which is in turn connected to a central router using a trunk link. The router will handle inter-VLAN communication using Router-on-a-Stick configuration.



Tasks:

1. On the access switch, create VLANs for each department (e.g., VLAN 10 for HR, VLAN 20 for Finance, VLAN 30 for IT). Assign the respective switch ports to the correct VLANs based on the department to which each PC belongs.
2. On the central router, configure sub-interfaces for each VLAN with appropriate IP addresses (default gateways for PCs). Enable trunking on the interface connected to the switch and ensure encapsulation is set to dot1Q.
3. Assign static IP addresses to each PC within the corresponding VLAN subnet. Test that PCs within the same VLAN can communicate with each other, but not with other VLANs before routing is enabled.
4. After configuring the router, test inter-VLAN communication by pinging from a PC in HR to a PC in Finance or IT. Troubleshoot any communication issues, verify the VLAN database, trunk status, and sub-interface configurations.

Task-03

Using the provided network topology, configure Virtual Local Area Networks (VLANs) to segment the network based on department or functional requirements.

